Task 3 – Expense Claim Approval System

1. Test Results

#	Test Scenario	Expected Result	Actual Result	Status
1	Valid Claim – Manager Approval	Claim is saved as Approved, and the employee receives an approval email.	Manager approved a valid claim, Google Sheets status changed to Approved, and the employee received an approval email.	Passed
2	Valid Claim – Manager Rejection	Claim is saved as Rejected, and the employee receives a rejection email.	Manager rejected the claim, Google Sheets status changed to Rejected, and the employee received a rejection email. A clear receipt was also successfully processed through this path.	Passed
3	Duplicate Receipt	Duplicate claim is rejected, employee is notified, and no new row is added.	Duplicate receipt was detected, employee received a notification, and no new row was added.	Passed
4	Missing Information	Invalid claim is rejected, and the employee receives a validation email.	Missing expense description was detected, employee received an invalid-claim email, and no new row was added.	Passed
5	Unreadable Receipt	AI identifies an unreadable receipt and rejects the claim.	Receipt was identified as unreadable, validation errors were generated, and no row or manager approval email was generated.	Passed
6	Technical Error	Error Handler sends a technical error notification.	Error Handler successfully sent a technical error notification email.	Passed

2. Hardest Problem Encountered and Solution

The hardest problem I encountered was handling unreadable or blurry receipt images. Initially, the Gemini AI model extracted incorrect or fabricated information from a blurry receipt and treated the expense claim as valid.

To solve this problem, I improved the Gemini prompt by introducing a receipt readability field. The AI was instructed to identify whether the receipt was readable and return null for information that could not be confidently extracted.

I also updated the Gemini parser Code node to preserve the readability result and modified the Validation Code node to reject receipts unless their readability status was explicitly marked as "Readable".

After implementing these changes, I tested the workflow again using a blurry receipt. The system correctly rejected the claim, sent an invalid-claim notification to the employee, and prevented the claim from being added to Google Sheets or forwarded to the manager.

This improvement made the expense claim system more reliable and reduced the risk of approving claims based on incorrect AI-extracted information.

3. Additional Testing Verification

A clear receipt was successfully processed by Gemini. The receipt readability validation passed, the duplicate check passed, and a new Pending record was added to Google Sheets.

The manager received the approval request and rejected the claim. The Google Sheets status was updated to Rejected, and the employee received the rejection notification.

This confirmed that the system correctly handles both readable and unreadable receipt images.